

problema 2:

1) $S \rightarrow 0A0 \mid 1B1 \mid BB$

$A \rightarrow C$

$B \rightarrow S \mid A$

$C \rightarrow S \mid \epsilon$

a) $S \rightarrow 0A0 \mid 00 \mid 1B1 \mid BB \mid B \mid \epsilon$

$A \rightarrow C$

$B \rightarrow S \mid A$

$C \rightarrow S$

b) $S \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB \mid \epsilon$

$A \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$

$B \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$

$C \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$

c) Todos producen. No se alcanza C

$S \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB \mid \epsilon$

$A \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$

$B \rightarrow 0A0 \mid 00 \mid 1B1 \mid 11 \mid BB$

d) $x \rightarrow 0 \quad y \rightarrow 1$

$M \rightarrow Ax \quad N \rightarrow By$

entonces:

$S \rightarrow XM \mid XX \mid YN \mid YY \mid BB \mid \epsilon$

$A \rightarrow XM \mid XX \mid YN \mid YY \mid BB$

$B \rightarrow XM \mid XX \mid YN \mid YY \mid BB$

$$M \rightarrow Ax$$

$$N \rightarrow By$$

$$x \rightarrow 0$$

$$y \rightarrow 1$$

$$2) \quad S \rightarrow aAa \mid bBb \mid \epsilon$$

$$A \rightarrow c \mid a$$

$$B \rightarrow c \mid b$$

$$C \rightarrow cDe \mid \epsilon$$

$$D \rightarrow A \mid B \mid ab$$

1) C available

A available por C

B available por C

D available

$$S \rightarrow aAa \mid aAa \mid bBb \mid bBb \mid \epsilon$$

$$A \rightarrow c \mid a$$

$$B \rightarrow c \mid b$$

$$C \rightarrow cDe \mid cDe \mid De \mid \epsilon$$

$$D \rightarrow A \mid B \mid ab$$

6)

Unarias $A \rightarrow C$

$$B \rightarrow C$$

$$D \rightarrow A$$

$$D \rightarrow B$$

$$S \rightarrow a A a | a a | b B b | b b | \epsilon$$

$$A \rightarrow a | C D E | C E | D E | \epsilon$$

$$B \rightarrow b | C D E | C E | D E | \epsilon$$

$$C \rightarrow C D E | C E | D E | \epsilon$$

$$D \rightarrow a b | a | b | C D E | C E | D E | \epsilon$$

$$c) S \rightarrow a A a | a a | b B b | b b | \epsilon$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$D \rightarrow a b | a | b$$

D no es alcanzable

$$S \rightarrow a A a | a a | b B b | b b | \epsilon$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$d) X \rightarrow A A \quad Y \rightarrow B B$$

$$S \rightarrow X A | A A | Y B | B B | \epsilon$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$X \rightarrow A A$$

$$Y \rightarrow B B$$

$$g) S \rightarrow A S A | a B$$

$$A \rightarrow B | S$$

$$B \rightarrow b | \epsilon$$

$$a) S \rightarrow A S A | A S | S A | S | a B | a$$

$$A \rightarrow B | S$$

$$B \rightarrow b$$

b)

$$S \rightarrow S$$

$$P \rightarrow S$$

$$P \rightarrow \emptyset$$

$$S \rightarrow ASA \mid AS \mid SA \mid aB \mid a$$

$$A \rightarrow b \mid ASA \mid AS \mid SA \mid aB \mid a$$

$$B \rightarrow b$$

c)

No hay

d)

$$X \rightarrow a$$

$$Y \rightarrow AS$$

$$S \rightarrow YAS \mid SA \mid XB \mid a$$

$$A \rightarrow b \mid YAS \mid SA \mid XB \mid a$$

$$B \rightarrow b$$